THE VIRTUAL MEET EXPERIENCE

2018 - 2019

HS VIRTUAL CHALLENGE MEET #4

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2018-2019 Virtual Challenge Meet #4 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.8.x**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

Which of the following is equivalent to the expression $11011101_2 + 335_8$?

A. 110111010_2 B. 12322_4 C. $1BA_{16}$ D. 442_{10} E. All are equivalent

QUESTION 2

What is output by the code to the right?

A. 12 B. 9
C. 17 D. 7
E. There is no output due to a runtime error.

```
out.println(4 + 5);
```

QUESTION 3

What is output by the code to the right?

A. false B. False
C. True D. false false
E. true

```
int j = 45 % 7;
int h = 7 % 45;
boolean f = h == j;
out.println( f );
```

QUESTION 4

What is output by the code to the right?

A. -8.5 B. 5.0
C. -9 D. 5
E. 8.0

```
double k=-8.5;
out.println(Math.min(Math.abs(k),5));
```

QUESTION 5

What is output by the code to the right?

A. ito B. mit
C. mio D. och
E. There is no output due to a syntax error.

```
String s = "mitochondria cell";
out.println(s.substring(0,2) +
                        s.charAt(3));
```

QUESTION 6

What is output by the code to the right?

A. True B. true
C. False D. false
E. None of the above.

```
boolean g = true;
boolean h = false;
out.println(!g&&(h|| g));
```

QUESTION 7

What is the output by the code to the right?

A. oes B. OES
C. banana sh D. BANANA SH
E. There is no output due to a runtime error

```
out.printf("%.9S%n", "banana shoes");
```

QUESTION 8

What is the output by the code to the right?

A. 212 B. 2

C. 0212 D. 12

E. There is no output due to a runtime error.

```
int y = 44;
switch(y) {
case 45 :
  out.print(0 + "");
case 44 :
  out.print(2 + "");
case 43 :
  out.print(1 + "");
case 42 :
  out.print(2 + "");
}
```

QUESTION 9

What is the output by the code to the right?

A. 9 B. 27

C. 8 D. 6

E. There is no output due to a syntax error.

```
int u = Integer.MAX_VALUE;
while(u > 10) {
  u = u/10;
}
out.println
(Integer.MAX_VALUE % 10 + u);
```

QUESTION 10

What is the output by the code to the right ?

A. Blenders B. 0

C. null D. Mist

E. There is no output due to an error.

```
String [] joe = new String[5];
joe[1] = "Kitchen";
joe[2] = "Spatula";
joe[3] = "With";
joe[4] = "Chicken";
joe = new String [6];
System.out.println(joe[2]);
```

QUESTION 11

What is the output by the code to the right ?

A. sci B. aplus

C. com D. comp

E. There is no output due to a runtime error.

```
String g = "aplus,comp,sci,com";
Scanner file = new Scanner(g);
file.useDelimiter(",");
file.next();
String s = file.next();
System.out.println(s);
```

QUESTION 12

What is the output by the code to the right ?

A. 10 B. 15

C. 12 D. 20

E. There is no output due to a runtime error

```
Integer x = 0;
for(int i = 0; i < 10; i+=2) {
    x += i;
}
System.out.print(x);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. II, IV, III, I B. IV, III, II, I

C. II, I, III, IV

D. I, II, III, IV

E. I, III, IV, II

```
I. ()
II. /=
III. +
IV. &(bitwise)
```

QUESTION 14

What is the output by the code to the right ?

A. 4 B. 3

C. 6 D. 5

E. There is no output due to a runtime error

```
int j = 4; int i = 0;
int [] sort = {1, 2 , 3, 3, 4, 4, 6,
7, 8, 9, 10};
for(i = 0; i < sort.length; i++) {
  if(sort[i] == j) {
     break;
  }
}
System.out.println(i);
```

QUESTION 15

What is the output by the code to the right ?

A. 24 B. 0

C. 88 D. 2

E. There is no output due to a runtime error.

```
ArrayList<Integer> a;
a = new ArrayList <Integer>();
a.add(99);
a.add(24);
a.add(88);
System.out.println(a.remove(2));
```

QUESTION 16

What is the output by the code to the right ?

A. 135 B. 126

C. 144 D. 180

E. There is no output due to a syntax error.

```
public int run(int l) {
  if(l <= 0) return l;
  return l + run (l - 9);
}
////////////CLIENT CODE/////////////
out.println(run(45));
```

QUESTION 17

What is the output by the code to the right ?

A. ejeurt B. truehje

C. true[h, e] D. [e, h]true

E. There is no output due to an error.

```
char[] ph = {'h', 'j', 'e'};
StringBuffer lo = new StringBuffer();
lo.append(true);
lo.append(ph);
lo.reverse();
lo.deleteCharAt(2);
out.println(lo);
```

QUESTION 18

What could replace **<1*>** in the code to the right, so that the code will work as intended?

A. <String,Integer>

B. <char, Integer>

C. <char,int>

D. <Character,Integer>

E. <Character, int>

QUESTION 19

Assuming that **<1*>** is filled in correctly, what is the output by the code to the right ?

A. 144 B. null

C. 166 D. 0

E. There is no output due to a runtime error.

```
TreeMap <1*> tm;
tm = new TreeMap<1*> ();
tm.put('l', 66);
tm.put('j', 78);
tm.replace('l', 88);
out.println
(tm.get('l') + tm.get('j'));
```

QUESTION 20

What is the value of 1011111_2 ?

A. 94_{10} B. 1233_4 C. $5F_{16}$ D. 136_8 E. 132_8

QUESTION 21

What is output by the code to the right ?

A. `ColeDrickarLogic`

B. `JKenLamar`

C. `ColeDrickar`

D. `ColeDrickLogic`

E. There is no output due to a runtime error.

```
String m;
m="J Cole Ken Drick Lam ar Logic";
Scanner sc=new Scanner(m);
while(sc.hasNext()) {
  if(sc.next().length()<4)
     out.print(sc.next());
}
out.println();
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right ?

A. `implements Comparable<Class>`

B. `implements Comparable`

C. Nothing is needed to replace **<1*>**

D. B or C

E. All of the above

QUESTION 23

Which of the following terms most accurately describes `Comparable`?

A. `abstract class` B. `class`

C. `interface` D. `child class`

E. None of the above.

QUESTION 24

What is the output by the line marked `//1` ?

A. `28chicken`

B. `7chicken7chicken7chicken7chicken`

C. `56chicken`

D. `7chicken`

E. There is no output due to an error.

QUESTION 25

What is the output by the line marked `//2` ?

A. `14` B. `7` C. `28`

D. There is no output due to a compile error.

E. There is no output due to a runtime error.

```
public class Class <1*>{
  private int j;
  private String s;
  public Class(int jj, String ss) {
     j = jj;
     s = ss;
  }
  public String go(int h) {
     return 4 * compareTo(h) + s;
  }
  public int compareTo(Object o) {
     int i = (int)(o);
     if(i < j) {
        return j - i;
     }
     return i - j;
  }
}
/////////client code////////////////
Class c = new Class( 7, "chicken");
out.println(c.go(14)); //1
out.println(c.j); //2
```

QUESTION 26

What is output by the code to the right?

A. `false true`

B. `false false`

C. `true false`

D. `true true`

E. There is no output due to an error

```
String h="SICKO MODE";
String b="Mo Bamba";
String g=".*\\s+.+";
out.print(h.matches(g)+" ");
out.println(b.equals(g));
```

QUESTION 27

What is output by the code to the right?

A. `0`

B. `6`

C. `9`

D. `12`

E. `16`

```
int[][]h=new int[5][5];
for(int r=0;r<5;r++)
  for(int c=0;c<5;c++)
          h[r][c]=(r+1)*(c+1);
out.println(h[2][3]);
```

QUESTION 28

What is output by the line marked `//q28`?

A. `96` B. `64`

C. `120` D. `81`

E. `24`

QUESTION 29

What is output by the line marked `//q29`?

A. `256` B. `120`

C. `24` D. `96`

E. `0`

QUESTION 30

What is output by the line marked `//q30`?

A. `5` B. `0`

C. `24` D. `120`

E. `1`

```
int recur(int y, int x) {
  if(x==1)
    return y;
  return x*recur(y,x-1);
}
///////client code//////////////////
out.println(recur(4,3)); //q28
out.println(recur(4,4)); //q29
out.println(recur(5,1)); //q30
```

QUESTION 31

What is the bit storage in the primitive type `double`?

A. `64` B. `8` C. `4` D. `16` E. `32`

QUESTION 32

What is output by the line marked `//1` code to the right?

A. `S`
B. `M`
C. `A`
D. `D`
E. There is no output due to an error.

QUESTION 33

What is output by the line marked `//2` code to the right?

A. `1`
B. `2`
C. `0`
D. `-1`
E. There is no output due to an error.

```
Stack<String>a=new Stack<String>();
a.push("S");
a.push("A");
a.push("D");
a.push("M");
a.push("A");
a.push("D");
a.pop();
a.pop();
out.println(a.pop());        //1
out.println(a.search("A")); //2
```

QUESTION 34

Which of the following data structures will keep all objects contained in natural sorted order according to the `Comparable` interface?

A. `Priority Queue`
B. `HashSet`
C. `TreeSet`
D. A and C.
E. All of the above.

QUESTION 35

What is output by the code segment to the right?

A. `[a, p, l, u, s, c, o, m, p, s, c, i]`
B. `apluscompsci`
C. Output cannot be determined until runtime.
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
char[]c="apluscompsci".toCharArray();
out.println(c);
```

QUESTION 36

What is output by the code segment to the right?

A. `O`
B. `E`
C. `true`
D. `false`
E. There is no output due to an error.

```
long g=256;
g+=6;
g/=2;
out.println(g%2==0?"O":"E");
```

QUESTION 37

What is output by the code segment to the right?

A. `[0, 3, 1, 0, 6, 0]`

B. `[0, 0, 0, 0, 0, 0]`

C. `[0, 0, 3, 1, 0, 6]`

D. There is no output due to a runtime error.

E. There is no output due to a compile error.

```
int[]k=new int[6];
k[1]=0;
k[2]=3;
k[3]=1;
k[4]=k[k[3]];
k[k[1]]=k[k[k[4]]];
k[5]=6;
out.println(Arrays.toString(k));
```

QUESTION 38

Which of the following 8-bit 2's complement binary numbers is equivalent to the decimal value -68?

A. 10111101

B. 01000011

C. 10111100

D. 01000100

E. 10111011

QUESTION 39

What is output by the code segment to the right?

A. `ZEZE ZEZE ZEZE`

B. `KEKE KEKE KEKE`

C. `KEZE ZEZE ZEZE`

D. `KEKE ZEKE KEKE`

E. There is no output due to an error.

```
String h="ZEZE ZEZE ZEZE";
h.replace("Z", "K");
out.println(h);
```

QUESTION 40

What is the inorder traversal of the binary search tree created by inserting the following values from left to right ? Assume duplicates are inserted on the left.

`12 2 4 2 90 212 17 2 8 -70 3 9 52 13 5`

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2018-2019 Virtual Challenge Meet #4
Computer Science - KEY

1) E
2) B
3) A
4) B
5) C
6) D
7) D
8) A
9) A
10) C

11) D
12) D
13) E
14) A
15) C
16) A
17) A
18) D
19) C
20) C

21) C
22) D
23) C
24) A
25) D
26) C
27) D
28) E
29) D
30) A

31) A
32) B
33) B
34) C
35) B
36) B
37) C
38) C
39) A
40) [-70, 2, 2, 2, 3, 4, 5, 8, 9, 12, 13, 17, 52, 90, 212]

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors.**
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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